HAM10000 Skin Lesion Classification

Image analysis and DenseNet121 transfer learning

Project type: Image data
Team: Individual project
Member: Luu Chi Cuong (2670811)

Problem and dataset

The project validates image-to-label matching, describes the HAM10000 dermatoscopic image collection, and trains a seven-class image classifier. The dataset contains 10,015 readable images, all 600 by 450 pixels, labeled with the target field `dx`: `akiec`, `bcc`, `bkl`, `df`, `mel`, `nv`, and `vasc`. The largest class, `nv`, represents about 67% of the images.

Split and training method

Data are split by `lesion_id` into 6,410 training images (64.0%), 1,608 validation images (16.1%), and 1,997 test images (19.9%). Group-aware splitting keeps images of one lesion in only one partition. DenseNet121 starts from ImageNet-pretrained weights; its classifier is warmed up before fine-tuning the full network. Images are resized with padding to 224 by 224. The training loss uses square-root inverse-frequency class weights, giving minority classes more influence without duplicating images. The checkpoint is selected on validation data using the average of accuracy and macro-F1.

Test results

Class	Precision	Recall	F1-score	Support
akiec	0.5672	0.5588	0.5630	68
bcc	0.6593	0.7059	0.6818	85
bkl	0.6151	0.7566	0.6786	226
df	0.7200	0.6429	0.6792	28
mel	0.6519	0.4703	0.5464	219
nv	0.9141	0.9154	0.9147	1,336
vasc	0.8500	0.9714	0.9067	35
Overall accuracy				82.47%
Macro-F1				71.01%
Balanced accuracy				71.73%

Interpretation and limitations

Class weighting mitigates the impact of imbalance during training, but the dataset and evaluation split remain naturally imbalanced. The difference between accuracy (82.47%) and macro-F1 (71.01%) reflects uneven class performance. Melanoma recall is the main weakness at 47.03%; 68 melanoma test images are predicted as nevi. The high vascular-lesion recall is based on only 35 test images and should be interpreted cautiously. These results are an academic benchmark, not a clinical diagnostic system.

Notebook and project details

Notebook: View on GitHub
Run in Colab: Open notebook in Google Colab
Dataset source: HAM10000 on Kaggle